History-Preserving Quadratic Normal Forms for Circular Coalescence

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This paper is Paper XLV (version 1.0) of the RIME program. It studies complete effective-fusion logs and literal endpoints under a fixed circular rotation-collapse action.

Abstract

A terminal placed partition does not determine the mergers that produced it. For a circular packet system generated by positive rotation and the adjacent collapse $d(1) = 0$, we characterize the complete effective-fusion logs and terminal placements jointly attainable from any initial placement of t disjoint named packets on n sites. Every admissible ordered cyclic-neighbor merger schedule and compatible literal endpoint can be jointly realized by a single chronological word. We give a representative of length at most $\frac{7}{2}n(n-1)$ preserving the complete log and endpoint, with explicit routing budgets. At terminal rank s, the joint objects number $(t!/s!)s\binom{n}{s}$. For a specified singleton-start all-right history, the exact minimum is $\frac{3}{2}n(n-1)-1$. An exact surplus identity decomposes every realizing word's additional length into nonnegative rotation slack, weighted carrier backslides, and idle defects. Zero surplus forces a unique literal chronological word. The same endpoint alone has a linear-length realization. Thus retaining the prescribed merger history can raise the cost of that endpoint from linear to quadratic.

Keywords: circular coalescence; fusion history; transformation semigroups; constructive normal forms; word length

1 Introduction

An execution of a merging system has several possible outputs. Its final transformation records where the atoms went. Its terminal placed partition records which atoms now travel together and where their packets are. A complete effective-fusion log additionally records the actual ordered parents of every merger and the order in which the mergers occurred. Executions with the same endpoint can have different logs, and prescribing a log can change the shortest realization substantially.

We consider the literal alphabet consisting of a positive rotation p on an n-cycle and one adjacent idempotent d, which sends site 1 to site 0 and fixes all other sites. Every occurrence of either letter has unit cost. The initial packets may be arbitrary nonempty, disjoint atom sets at arbitrary distinct sites. Their initial physical cyclic order, rather than a numerical order on the atoms, supplies the relevant geometry.

Our construction replays an ordered merger schedule on the placement left by each preceding step, then aligns the endpoint by fusion-free gap flow. Boundary deletions enumerate the resulting log-endpoint pairs. For one all-right history, rotation and hole ledgers account for every excess letter and force the unique shortest chronological word.

These questions are adjacent to the study of transformation semigroups generated by a group and an idempotent [5], and generating ranks for endomorphisms of finite circular orders [4]. Cameron, Castillo-Ramirez, Gadouleau and Mitchell study $\ell(D, \alpha)$, the minimum number of digraph-arc generators expressing a transformation α [1, Section 1]. Each available arc $a \to b$ is a defect-one idempotent letter of unit cost. Here only the fixed letters p, d have unit cost. For

example, the circular adjacent arc $b+1 \to b$ is realized chronologically by $p^{[-b]_n} d p^{[b]_n}$, whose expanded length is $1 + [-b]_n + [b]_n$: it is $n+1$ when $b \neq 0$, and one when $b = 0$. The displayed conjugate is not an additional unit-cost letter.

Their target is an endpoint transformation. From our singleton start, the literal endpoint determines the induced transformation on atoms, but equality of the complete log and endpoint is strictly finer than endpoint equality alone, as Section 7 demonstrates. More generally, our target from any fixed packet start P_0 is the pair consisting of its complete ordered fusion log and its terminal placed partition.

2 Placements and Complete Logs

Fix $n \geq 3$ and $Q = \mathbb{Z}/n\mathbb{Z}$. Let Ω be a finite nonempty atom set, independent of Q. A placed partition is a map $P : Q \to 2^{\Omega}$ whose nonempty values are pairwise disjoint and cover Ω. Its occupied sites and rank are

$$\operatorname{supp} P = \{q : P(q) \neq \emptyset\}, \qquad \operatorname{rank} P = |\operatorname{supp} P|. \tag{2.1}$$

The nonempty values are packets, distinguished by their literal atom sets. The site maps are

$$p(q) = q + 1, \qquad d(1) = 0, \quad d(q) = q \quad (q \neq 1). \tag{2.2}$$

They act by pushforward union:

$$(a_{\#}P)(y) = \bigcup_{q : a(q) = y} P(q), \qquad a \in \{p, d\}. \tag{2.3}$$

Words are chronological: uv means execute u and then v. Write $P \cdot w$ for the successor. A **fusion** occurs exactly when a d step has both sites $0, 1$ occupied. Its complete record is

$$\rho = (d, 0, 1, U, V, 0, U \cup V), \qquad U = P(0), \quad V = P(1). \tag{2.4}$$

The order of U, V is part of the record. Let $\mathcal{H}_P(w)$ be the list of all such records in chronological order. Nonfusing letters add nothing to the list. Every fusion reduces rank by one, while every other step preserves rank. Thus an execution from rank t to rank s has exactly $t - s$ recorded events.

For an integer a, let $[a]_n$ be its least nonnegative residue. A power p^k denotes k charged applications of p.

Fix an initial placement P_0 of rank t. List its packets in positive physical cyclic order as $B_0, \dots, B_{t-1}$, with indices understood cyclically. A cyclic block of initial packets is a union of consecutive B_i in that order. No numerical cyclic order on Ω is required.

Lemma 2.1 (cyclic packet order)**.**
In every execution from P_0, each packet is a cyclic block of initial packets, and the physical cyclic order of the current packets agrees with their induced cyclic order. Every effective merger combines ordered cyclic neighbors. At rank two both ordered pairs are possible cyclic neighbor pairs.

Proof. *The assertion holds initially. A p step rotates the physical positions and changes no packet. A nonfusing d either fixes the placement or moves the packet at site 1 into the adjacent empty site 0. It crosses no occupied site and preserves cyclic packet order. A fusing d replaces the packets at adjacent sites $0, 1$ by their union. They are cyclic neighbors in that ordered direction, so their union is again a cyclic block and the remaining packet order is preserved. Induction on the word proves the assertion. When only two packets remain, each is the next packet after the other in the cyclic list.*

*An **admissible schedule** from P_0 is a finite sequence of records of the form (2.4), constructed by repeatedly choosing two distinct ordered cyclic neighbors in the current packet list and replacing them by their union. At rank one no further merger is allowed. The initial list is the list of packets of P_0. For a schedule ending at rank s, a **compatible endpoint** is any placement of its final s packets at distinct sites in the induced cyclic order. Admissibility and compatibility are defined directly from this packet cycle.*

3 Fusion-Free Routing

Lemma 3.1 (a batch through consecutive holes)**.**

Suppose a packet V is at site b, and the sites $b-1, \dots, b-f$ are holes, where $1 \le f \le n-2$. The word

$$\mathsf{B}_{b,f} = p^{[1-b]_n} d (pd)^{f-1} p^{[b-f]_n} \tag{3.1}$$

moves V to $b-f$, restores every other packet to its original site, and produces no fusion. Its length is at most $2n+f$.

Proof. *The prefix sends V to site 1 and the preceding original hole $b-1$ to site 0. The first d moves V to 0. Each subsequent p returns V to site 1, and the next d moves it into the next preceding hole. Before the kth d, the site at port 0 corresponds to the original site $b-k$, for $1 \le k \le f$. Thus no one of these steps fuses packets or moves another packet by d.*

All other packets receive the total rotation

$$[1-b]_n + (f-1) + [b-f]_n \equiv 0 \pmod{n}.$$

They therefore return to their starting sites. The last d leaves V at 0, and the suffix puts it at $b-f$. If $f \ge 2$, the two outer exponents sum to a nonnegative integer congruent to $1-f$ modulo n and at most $2n-2$; hence $[1-b]_n + [b-f]_n \le 2n+1-f$. Adding the $2f-1$ intervening letters gives length at most $2n+f$. If $f = 1$, the outer exponents sum to 0 or n, giving length 1 or $n+1$.

A nonempty use of (3.1) is called one batch and accounts for f unit hole transfers.

Lemma 3.2 (one ordered merger)**.**

Let U, V be ordered cyclic neighbors at sites a, b of a placement with at least two packets. Set $f = [b-a-1]_n$. There is a word that produces exactly the event $(d, 0, 1, U, V, 0, U \cup V)$, with no earlier fusion, using at most one batch, f unit hole transfers and $3n+f$ letters.

Proof. *All f sites between a and b are holes. If $f > 0$, apply Lemma 3.1 to V; the rank assumption ensures $f \le n-2$. It moves V to $a+1$ and restores every other packet, including U. For $f = 0$ no batch is needed. Rotate by $p^{[-a]_n}$ and execute one d. The parents now occupy ports $0, 1$ in the requested order, so the event is exactly the specified tuple. The batch costs at most $2n+f$, and the final rotation and fusion cost at most n. When only two packets remain, apply this construction separately to (U, V) and (V, U). Both are ordered cyclic neighbors, and the two resulting records remain distinct even though their result sets coincide.*

Proposition 3.3 (same-order placement)**.**

Suppose $P, P^\star$ have the same s named packets in the same cyclic order. A fusion-free word takes P to $P^\star$; when $s = 1$, it is a rotation. Its batch and hole-transfer budgets, and its letter bound, are

$$K_{\text{place}} \le s-1, \qquad M_{\text{place}} \le (n-s)(s-1), \tag{3.2}$$

and

$$H(n,s) = (3n-s)(s-1) + (n-1) \tag{3.3}$$

Proof. *Handle $s = 1$ by rotation. For $s \geq 2$, fix a cyclic naming $C_0, \dots, C_{s-1}$ of the packets. Let g_i and $g_i^\star$ be the numbers of holes after C_i and before C_{i+1} in the two placements. Both vectors sum to $h = n - s$.*

Choose nonnegative integral flows satisfying

$$g_i^\star = g_i + f_{i-1} - f_i. \tag{3.4}$$

For example, form successive prefix sums of $g_i - g_i^\star$ around the cycle and subtract their minimum. Their total increment is zero, so this gives (3.4) and at least one zero-flow edge b. Moreover $f_i \leq h$: relative to an edge where the minimum occurs, f_i is the difference of the current and target gap masses on one cyclic interval, each between 0 and h.

Starting after b, process edges $b+1, \dots, b+s-1$ in cyclic order. At edge i move C_{i+1} backward through f_i holes by Lemma 3.1, omitting the step when $f_i = 0$. The preceding transfer has added f_{i-1} holes to gap i; later transfers have not yet touched it. Its current size is therefore

$$g_i + f_{i-1} = g_i^\star + f_i \geq f_i. \tag{3.5}$$

Every batch is enabled on the actual placement left by its predecessor. It decreases gap i by f_i and increases gap $i+1$ by that amount, without moving another packet or creating a fusion. When the last edge has been processed, (3.4) also fixes the zero-flow gap b, so all gaps equal their target values.

There are at most $s-1$ nonzero flows and their sum is at most $h(s-1)$. The batches cost at most $2n(s-1) + h(s-1)$. A final rotation sends C_0 to its prescribed site, costing at most $n-1$ letters. Equal gap vectors then fix every other named packet at its prescribed site as well. This proves literal placement equality and (3.2)–(3.3). If $s = n$, both gap vectors and every normalized flow are zero, so the construction uses only the final rotation.

4 Joint Realization and Quadratic Normal Form

Theorem 4.1 (log and endpoint realization)**.**

Fix any initial placement P_0 of rank t, and let $1 \leq s \leq t \leq n$. A structured list $\mathcal{H}$ and a placement $P^\star$ of rank s are jointly attainable from P_0 if and only if $\mathcal{H}$ is an admissible schedule and $P^\star$ is compatible with it. When these conditions hold, one word W satisfies

$$\mathcal{H}_{P_0}(W) = \mathcal{H}, \qquad P_0 \cdot W = P^\star, \tag{4.1}$$

with

$$\begin{aligned} |W| \leq B(n,t,s) := 3n(t-s) + (n-t)(t-s) + \binom{t-s}{2} \\ + (3n-s)(s-1) + (n-1). \end{aligned} \tag{4.2}$$

Proof. *Necessity follows from Lemma 2.1 and the exact event definition. For sufficiency, let $d = t - s$ be the number of schedule records and write $\mathcal{H} = (\rho_1, \dots, \rho_d)$. Starting at P_0, replay the records by Lemma 3.2. After j events maintain the invariant that the actual log is $(\rho_1, \dots, \rho_j)$ and the actual source-block partition is the abstract schedule partition after those events. Physical positions are not part of this induction invariant.*

The next recorded parents are present by that partition equality. Their ordered adjacency in the abstract cyclic list is also their ordered adjacency in the actual placement, by Lemma 2.1. Lemma 3.2 therefore applies to the current concrete placement, appending exactly the next record and making the required union. The induction continues from the placement reached by the preceding record.

After the d events, the current and target placements have the same packets in the same cyclic order. Proposition 3.3 restores $P^\star$ without adding an event. The concatenated word proves both equalities in (4.1). If $t = s$, the replay stage is empty and only Proposition 3.3 is used. If $s = 1$, its final stage is just the rotation of the single packet, including the case $t = s = 1$.

Initially there are $h = n - t$ holes. Before event $j+1$ there are $h + j$ holes, so its parent gap has length $f_j \le h + j$. Its route costs at most $3n + f_j$. Thus the complete fusion stage costs at most

$$\sum_{j=0}^{d-1}(3n + f_j) \le 3nd + hd + \binom{d}{2}. \tag{4.3}$$

Adding $H(n, s)$ proves (4.2), including an empty schedule.

Corollary 4.2 (uniform bound and routing budgets)**.**
The word in Theorem 4.1 can be chosen with

$$|W| \le \frac{7}{2}n(n-1), \qquad K \le t - 1, \tag{4.4}$$

and

$$M \le (t-s)(n-t) + \binom{t-s}{2} + (n-s)(s-1). \tag{4.5}$$

For a fully occupied start, $K \le n - 2$ when $s < n$; when $s = n$, a rotation of at most $n - 1$ letters suffices.

Proof. *The replay uses at most $t - s$ batches and the final placement at most $s - 1$, proving the bound on K. Summing the parent gaps in (4.3) and the terminal transfer bound proves (4.5).*

Define

$$Q(n, s) = 3n(n - s) + \binom{n-s}{2} + H(n, s).$$

With $h = n - t$, expansion gives

$$Q(n, s) - B(n, t, s) = 3nh + \binom{h}{2} \ge 0, \qquad Q(n, s+1) - Q(n, s) = -(n + s - 1) < 0. \tag{4.6}$$

It follows that $B(n, t, s) \le Q(n, 1) = \frac{7}{2}n(n-1)$. If $t = n$ and $s < n$, the first merger has zero parent gap, removing one batch from the bound. If $t = s = n$, the compatible endpoint is a rotation of the start. Sparse zero-merger targets are covered by Proposition 3.3.

For the fully occupied start, the transfer budget also has the form

$$M \le \binom{n-s}{2} + (n-s)(s-1) = \binom{n-1}{2} - \binom{s-1}{2}.$$

The dependence of the length budget on terminal rank can equivalently be written

$$\frac{7}{2}n(n-1) - Q(n, s) = \frac{(s-1)(s+2n-2)}{2}.$$

These are identities for the construction's budget, not formulas for the shortest length of each target. In the zero-fusion fully occupied case, $K = M = 0$ and the sharper length bound is $n - 1$.

Corollary 4.3 (complete effective-log compression)**.**

For every word w from P_0, there is a word W of length at most $B(n,t,s)$, where $s = \operatorname{rank}(P_0 \cdot w)$, with

$$(\mathcal{H}_{P_0}(W), P_0 \cdot W) = (\mathcal{H}_{P_0}(w), P_0 \cdot w). \tag{4.7}$$

Proof. *Its actual log is admissible and its endpoint compatible, so apply Theorem 4.1.*

Every observation determined by the complete effective log and literal endpoint in (4.7) is preserved.

5 Log Counts and Forgetful Fibres

For $t = 1$ the only schedule is empty. For $t \geq 2$, mark the t original directed boundaries between successive B_i. Until rank one is reached, every current packet corresponds to one interval of this initial packet circle. Merging ordered neighbors deletes the original boundary from the left parent to the right parent.

Lemma 5.1 (boundary-deletion encoding)**.**

Schedules ending at rank s are in bijection with ordered lists of $t-s$ distinct initial boundaries.

Proof. *With at least two current intervals, each remaining directed boundary has a distinct interval on either side. Deleting it merges those two intervals in the boundary's order, so every ordered list of $t - s$ distinct boundaries gives a schedule.*

For the inverse, retain the initial boundary labels on the current interval circle. The complete ordered parents in the next record identify the unique boundary directed from the last initial packet of the left interval to the first initial packet of the right interval. Delete this label and continue. At rank two the two possible parent orders select the two different remaining boundaries, although both result sets are the full atom set. Induction on the record list proves that encoding and decoding are inverse. At rank one the last undeleted boundary is not visible in the endpoint, but is recovered as the unique label absent from the deletion list. The $t = s = 1$ case is the unique empty list.

Theorem 5.2 (rank-refined joint count)**.**

From any fixed rank-t start, for $1 \leq s \leq t$,

$$\#\{\mathcal{H} : \text{terminal rank } s\} = \frac{t!}{s!}, \qquad \#\{P^\star \mid \mathcal{H}\} = s\binom{n}{s}, \tag{5.1}$$

and therefore

$$\#\{(\mathcal{H}, P^\star) : \operatorname{rank} P^\star = s\} = \frac{t!}{s!}\, s\binom{n}{s}. \tag{5.2}$$

Proof. *Lemma 5.1 gives $t!/s!$ boundary lists. Fix one log and its s resulting named packets. Choose an s-element occupied-site set in $\binom{n}{s}$ ways and choose which packet occupies its least site in s ways. The cyclic order fixes every other packet at every other occupied site. Conversely the literal placement recovers both choices. Theorem 4.1 realizes each choice for this very log. This proves both fibre cardinality and the product count. For $s = 1$ the occupied-site choice is one of the n phases.*

More explicitly, let $\mathsf{Joint}_{P_0,s}$ consist of actual complete logs paired with their literal endpoints, rather than execution words. Write $\mathsf{Logs}_{t,s}$ for the admissible logs, $\mathsf{End}(\mathcal{H})$ for the literal endpoints jointly attainable with one such log, and $\mathsf{Coord}_{n,s}$ for occupied-site and packet-anchor choices. The argument establishes the dependent bijections

$$\mathsf{Joint}_{P_0,s} \simeq \coprod_{\mathcal{H} \in \mathsf{Logs}_{t,s}} \mathsf{End}(\mathcal{H}) \simeq \mathsf{Logs}_{t,s} \times \mathsf{Coord}_{n,s}.$$

The first map retains the log and every packet at its physical site. For the second, fix the terminal packet cycle computed from the log: encoding reads its occupied sites and anchor, while decoding places that same packet cycle at those sites. Both composites recover the literal joint object.

Proposition 5.3 (fibres over literal endpoints)**.**
At rank $s \geq 2$ there are

$$s\binom{n}{s}\binom{t}{s} \tag{5.3}$$

attainable literal endpoints. Every such endpoint admits exactly $(t-s)!$ complete logs. At rank one there are n endpoints, each admitting $t!$ complete logs.

Proof. *If $s \geq 2$, the endpoint's source partition determines its s surviving initial boundaries. The remaining $t-s$ boundaries can be deleted in any order; Lemma 5.1 and Theorem 4.1 make these exactly the logs over that endpoint. Choosing the surviving boundaries gives $\binom{t}{s}$ source partitions, and each has $s\binom{n}{s}$ placements. At rank one no surviving boundary is identified by the full atom set. There are t choices of that boundary and $(t-1)!$ deletion orders, giving $t!$ logs over each endpoint. This also covers $t = 1$.*

The coordinate-only fibre in (5.1) forgets packet identities and has $t!/s!$ logs; Proposition 5.3 instead fixes a literal endpoint.

For a singleton start $t = n$, summing (5.2) over rank gives

$$n!\sum_{s=1}^{n}\frac{\binom{n}{s}}{(s-1)!}. \tag{5.4}$$

The rank-one contribution alone is $n\,n!$. Equation (5.4) counts distinct complete logs paired with literal endpoints.

6 A Specified Quadratic Optimum

Now take $\Omega = Q$ and the singleton start $P_{\text{seed}}(q) = \{q\}$. Prescribe the all-right log

$$\mathcal{H}_R = (\rho_1, \dots, \rho_{n-1}), \qquad \rho_i = (d, 0, 1, \{0, \dots, i-1\}, \{i\}, 0, \{0, \dots, i\}), \tag{6.1}$$

and prescribe the final packet Q at site $n-1$. Let $L_R(n)$ be the minimum length realizing this log and endpoint.

Lemma 6.1 (an explicit realization)**.**
The chronological word

$$W_n = d\left[\prod_{j=1}^{n-2} p^{n-j} d (pd)^j\right] p^{n-1} \tag{6.2}$$

realizes the target and has length $\frac{3}{2}n(n-1) - 1$.

Proof. *The initial d creates $\{0, 1\}$ at 0. Before stage j, the carrier is $\{0, \dots, j\}$ at 0, the sites $1, \dots, j$ are holes, and the remaining singletons occupy their original sites. After p^{n-j}, singleton $j+1$ is at 1 and site 0 is empty. The first d slides that singleton to 0. The next j copies of pd slide it through the holes; on the last copy the carrier has returned to 0, so the d fuses the requested ordered parents. There are j nonfusing slides and one fusion in this stage. Every other packet receives a full n rotations and no slide, so its original site is restored. This proves the stage invariant.*

After the last stage the full packet is at 0, and the suffix puts it at $n-1$. The stages cost $n+j+1$ letters each, so

$$|W_n| = 1 + \sum_{j=1}^{n-2}(n+j+1) + (n-1) = \frac{3}{2}n(n-1) - 1. \tag{6.3}$$

The number of rotations is $P_{\min} = n(n-1) - 1$*, and the number of nonfusing slides is* $T = (n-1)(n-2)/2$*.*

We prove optimality by accounting separately for rotations and slides. Because (6.1) prescribes the complete fusion log, every realizing word has exactly those $n-1$ *fusions: no additional outside-only fusion is allowed. Consequently every packet not yet absorbed into the carrier remains its original singleton. Every other* d *step is then an external-singleton slide, a carrier backslide from* 1 *to* 0*, or a step that leaves the placement unchanged. Write their counts as* S, B, Z*, respectively. Let* B_m *count carrier backslides while exactly* m *external singletons remain; thus* $B = \sum_{m=0}^{n-2} B_m$*.*

Lemma 6.2 (rotation cost of a backslide)**.**

For any realizing word w*,*

$$N_p(w) \geq P_{\min} + B. \tag{6.4}$$

Proof. *Before the first fusion, the placement is fully occupied, so the prefix consists of rotations. The first ordered parents in (6.1) force their number to be a multiple of* n*.*

Between consecutive specified fusions, the carrier starts and ends at 0*. Each rotation increments its site and each backslide decrements it, so* $N_p - B$ *on that interval is a multiple of* n*. Immediately after a fusion, and immediately after a carrier backslide, the carrier is at* 0 *and site* 1 *is empty. At least one rotation is needed before the next backslide. After the last backslide, or after the preceding fusion if there was no backslide, at least one further rotation is needed before the next specified fusion. A* d *cannot populate the empty site* 1*. Consequently* $N_p \geq B+1$ *on the interval, making the multiple of* n *strictly positive. It follows that* $N_p \geq n + B$ *on each of the* $n-2$ *inter-fusion intervals.*

After the last fusion only the carrier remains. The same site accounting gives $N_p - B \equiv n-1 \pmod{n}$ *on the tail. Every backslide requires a preceding rotation, so this difference is nonnegative and hence at least* $n-1$*. Adding the intervals proves (6.4).*

Lemma 6.3 (hole-potential identity)**.**

The external slides satisfy

$$S = T + \sum_{m=1}^{n-2} mB_m, \qquad T = \frac{(n-1)(n-2)}{2}. \tag{6.5}$$

Proof. *While* m *external singletons remain, anchor their cyclic order at the carrier and let* $g_0, \ldots, g_m$ *be the successive hole gaps. In particular* g_0 *follows the carrier. Set*

$$\Phi_m(g) = \sum_{i=0}^{m}(m-i)g_i. \tag{6.6}$$

Rotations and unchanged d *steps preserve this potential. An external singleton slide transfers a hole from its preceding gap to its following gap, decreasing* Φ_m *by one. A carrier backslide transfers a hole from* g_m*, whose coefficient is zero, to* g_0*, whose coefficient is* m*, increasing the potential by* m*.*

At a prescribed right fusion the gap g_0 *is zero. After the fusion the new gaps are* $(g_1 + 1, g_2, \ldots, g_m)$*: the vacated site* 1 *adds one hole immediately after the enlarged carrier. Therefore*

$$\Phi_{m-1}(g_1+1, g_2, \ldots, g_m) - \Phi_m(0, g_1, \ldots, g_m) = m - 1. \tag{6.7}$$

Immediately after registration $m = n-2$, with gap vector $(1, 0, \ldots, 0)$ and potential $n-2$. At rank one the potential is zero. Summing all changes yields

$$0 = (n-2) - S + \sum_{m=1}^{n-2} mB_m + \sum_{m=1}^{n-2} (m-1).$$

The two terms independent of backslides sum to T, proving (6.5). The rank-one backslides B_0 have no effect on this identity.

Theorem 6.4 (exact optimum and excess cost)**.**

The specified target has exact minimum

$$L_R(n) = \frac{3}{2}n(n-1) - 1. \tag{6.8}$$

More precisely, put $R = N_p(w) - P_{\min} - B \geq 0$. Every realizing word satisfies

$$|w| - L_R(n) = R + 2B_0 + \sum_{m=1}^{n-2} (m+2)B_m + Z. \tag{6.9}$$

Proof. *There are $n-1$ fusing letters, so $|w| = N_p + (n-1) + S + B + Z$. Substitute (6.4)–(6.5) and $N_p = P_{\min} + B + R$. The constant part is $P_{\min} + (n-1) + T = \frac{3}{2}n(n-1) - 1$, and the remaining terms give (6.9). They are all nonnegative. Lemma 6.1 attains the bound.*

Since every term in (6.9) is nonnegative, equality forces $R = Z = B_m = 0$ for every m. Appendix A derives the literal word from these equalities.

Corollary 6.5 (uniqueness of the shortest word)**.**

The only shortest word for this target is W_n in (6.2).

Equivalently, with RealizesAllR(W) *denoting exactly the complete log (6.1) and the specified literal endpoint,*

$$\exists! W \in \{p, d\}^*, \qquad \text{RealizesAllR}(W) \wedge |W| = \frac{3n(n-1)}{2} - 1.$$

Proof. *Appendix A shows that every zero-surplus realization equals W_n as a literal chronological word; Lemma 6.1 supplies existence.*

Let $D_n^{\log}$ be the maximum, over all jointly attainable complete logs and literal endpoints from the singleton start, of their minimum realization length. Theorem 5.2 makes this a maximum over a finite set. Theorems 4.1 and 6.4 give

$$\frac{3}{2}n(n-1) - 1 \leq D_n^{\log} \leq \frac{7}{2}n(n-1). \tag{6.10}$$

Thus $D_n^{\log}$ has quadratic order. The exact minimum $L_R(n)$ concerns the specified complete log and literal endpoint; $D_n^{\log}$ ranges over such targets. Neither quantity is the diameter of the underlying state transition graph.

7 Forgetting History Can Change the Cost Order

Consider the much shorter word

$$V_n = d(pd)^{n-2}p^{n-1}. \tag{7.1}$$

After the first d, every pd brings the preceding singleton to port 0 and the carrier to port 1, then fuses them. It reaches the full packet at 0 and finally at $n-1$. Hence it has exactly the same literal endpoint as W_n, with length $3n-4$. Its ordered merger log differs from (6.1).

Write E_n for the minimum length reaching this endpoint when history is unrestricted. A rank-n start requires $n-1$ effective mergers to reach rank one, so $n-1 \le E_n \le 3n-4$ and $E_n = \Theta(n)$. Meanwhile Theorem 6.4 implies

$$\frac{L_R(n)}{E_n} \ge \frac{\frac{3}{2}n(n-1)-1}{3n-4} \sim \frac{n}{2}. \tag{7.2}$$

This is an unbounded distortion under the map that forgets the complete log. It compares two optimization problems in the same literal alphabet from the same source, with the same terminal placement. The quadratic cost belongs to the prescribed chronological mergers, rather than to that endpoint alone.

8 Registered Observations

A registered observation can be read from a complete log by fixing an atom set F, retaining the first event whose result is exactly F as its origin, and then retaining later events in which the current packet containing F is a parent. The final placed partition supplies the current carrier and all outside packets. This recovers the origin and ordered selected absorption chain used in registered-summary classifications [2, 3].

Corollary 8.1 (compression of attainable registered summaries)**.**

Every registered summary attained from a singleton start has a word realization of length at most $\frac{7}{2}n(n-1)$ preserving its full origin, selected chain and literal terminal partition.

Proof. *Apply Corollary 4.3 to one word attaining the summary. Its complete log and endpoint determine all the specified reads.*

The corollary applies to attained summaries, including attained full-arc registrations; it does not classify which summaries are attainable.

9 Computational Artifacts

At the `paper45-v1.0` release tag, the Paper XLV companion index maps Theorems 4.1, 5.2 and 6.4, Proposition 5.3 and Corollary 6.5 to Lean declarations under `experiments/fusion_log_compression/`. Its `FORMALIZATION_MAP.md` gives exact declaration correspondences, while its `README.md` gives read-only reproduction commands and the scope of bounded finite-state checks.

10 Conclusion

The route construction and boundary-deletion encoding yield a dependent bijection of attainable complete logs and literal endpoints, with a quadratic normal form. For the specified all-right target, (6.9) accounts for every excess letter, and zero surplus forces the unique raw word W_n. Forgetting the log permits the same endpoint in linear length.

Appendix A Forcing the Unique Shortest Word

A.1 Activity and Exact Rotation Budgets

Call a defect letter **active** when its predecessor has port 1 occupied. If port 1 is empty, d leaves the full execution state unchanged and appends no effective event. Deleting that letter therefore preserves the complete log and literal endpoint. A minimum-length realization cannot contain such a letter: otherwise deletion contradicts the raw lower bound in Theorem 6.4.

Let w attain equality in (6.8). By (6.9), every B_m, R and Z vanishes, so $N_p(w) = P_{\min}$. Every actual inter-fusion interval contributes at least n rotations, and the tail at least $n-1$, by

Lemma 6.2. These lower bounds already sum to $P_{\min}$. Consequently the registration prefix has no rotations, every inter-fusion interval has exactly n, and the tail has exactly $n-1$. Before the first fusion a fully occupied placement permits no nonfusing d, so registration is the literal first letter d.

A.2 Local Stage Rigidity

Let $1 \le j \le n-2$. Suppose a stage begins with the actual placement

$$C_j = \{0, \dots, j\} \text{ at } 0, \quad 1, \dots, j \text{ empty}, \quad \{i\} \text{ at } i \quad (j+1 \le i < n). \tag{A.1}$$

Assume that its next effective fusion has ordered parents $(C_j, \{j+1\})$, its prefix before that fusion is fusion-free, every d in the stage is active, and it uses exactly n rotations. Then the entire stage, including the fusion letter, is

$$p^{n-j} d (pd)^j. \tag{A.2}$$

Proof. There is a defect before the final fusion: without one the n rotations restore (A.1), where port 1 is empty. Before the first defect only rotations have occurred. That defect is active and nonfusing, so port 0 is a hole and port 1 occupied. The only hole-to-packet boundary of (A.1) in the positive direction ends at singleton $j+1$. Among rotation counts from 0 to n, its ordered placement at ports $0, 1$ occurs exactly at $n-j$. This identifies the first defect as a slide of that singleton. The calculation is cyclic and includes the boundary crossing $n-1 \to 0$.

After this slide mark atom $j+1$. It is at port 0, with exactly j rotations remaining. Every other atom is at an integer representative k satisfying $2 \le k \le n-j$: the carrier is at $n-j$, and all other outside singletons are between 2 and $n-j-1$. During the remaining rotations their representatives stay in $[2, n]$, so none can reach port 1. A defect fixes all these atoms. Induction on the actual remaining letters therefore proves that every active defect moves the marked atom from 1 to 0; this is not an assumed slide ordering.

Lift the marked position to height n immediately after the first slide. Each remaining p raises the height by one, and each active d lowers it by one when its residue is 1. The height cannot fall below n and cannot reach $2n$, since fewer than n rotations remain. The prescribed final fusion places the marked atom at 0, so its terminal height is exactly n. Thus the remaining j rotations are paired with exactly j marked defects. Starting at height n, the next defect requires one rotation. Two rotations before it would pass height $n+1$, and the remaining budget cannot reach another height with residue 1. Hence each rotation must be followed immediately by d, returning the height to n. Repeating forces the suffix $(pd)^j$ and proves (A.2).

A.3 Induction over the Remaining Actual Log

After registration, the actual placement is C_1 in (A.1). Assume inductively that the next unprocessed stage begins at C_j. Cut the remaining raw word immediately after its next effective fusion. The complete log (6.1) fixes the ordered parents of that event, and no earlier fusion occurs in the extracted stage. By A.1 every intervening defect is active. Lemma 6.2 gives at least n rotations in this interval, the same bound for each later inter-fusion interval, and at least $n-1$ for the final tail. Since the entire word contains exactly $P_{\min} = n(n-2) + (n-1)$ rotations, equality holds in every separate bound. In particular, the current stage contains exactly n rotations.

The local argument A.2 now forces its entire literal word to be $p^{n-j} d (pd)^j$. Lemma 6.1 evaluates this word on C_j and gives precisely C_{j+1}, together with the next prescribed log event. This re-establishes the induction invariant on the actual execution left by the stage. Iterating to $j = n-2$ fixes all inter-fusion stages.

A.4 The Rank-One Tail and Unique Existence

After the last fusion the full packet is at 0. All defects in the remaining tail are active, so each would be a carrier return from 1 to 0. Starting at lifted height zero, the $n-1$ rotations give terminal height at most $n-1$. Its prescribed residue $n-1$ forces that height to equal $n-1$, leaving no downward step. Therefore the tail contains no d and is exactly p^{n-1}. This argument includes rank-one backslides even though their hole-potential weight is zero.

Combining registration, A.3 and the tail proves literal equality $w = W_n$. Together with the actual execution and length of W_n, this gives the unique-existence statement of Corollary 6.5.

References

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